

Detailed Solutions for Tutorial-1

MTL106/MTL2006: Probability and Stochastic Processes

Q1. Let $\Omega = \{0, 1, 2, \dots\}$. Let $\mathcal{F}$ be the collection of subsets of Ω that are either finite or whose complement is finite. Is $\mathcal{F}$ a σ -field? Justify your answer.

Solution:

A σ -field must satisfy three properties: it must contain the sample space Ω , be closed under complementation, and be closed under countable unions.

Let us test if $\mathcal{F}$ is closed under countable unions. Consider the singleton sets $A_n = \{2n\}$ for $n \in \{0, 1, 2, \dots\}$. Since each A_n contains exactly 1 element, it is finite, and therefore $A_n \in \mathcal{F}$ for all n .

Now, let A be the countable union of all these sets:

$$A = \bigcup_{n=0}^{\infty} A_n = \{0, 2, 4, 6, \dots\}$$

The set A is the set of all non-negative even integers, which is infinite. The complement of A in Ω is:

$$A^c = \Omega \setminus A = \{1, 3, 5, 7, \dots\}$$

The set A^c is the set of all positive odd integers, which is also infinite.

Since neither A nor A^c is finite, $A \notin \mathcal{F}$. Thus, $\mathcal{F}$ is not closed under countable unions. Therefore, $\mathcal{F}$ is **not a σ -field** (though it is an algebra).

Q2. Let $\mathcal{F}$ be a σ -field on Ω and let $B \in \mathcal{F}$. Show that $\mathcal{G} = \{A \cap B : A \in \mathcal{F}\}$ is a σ -field on B . Is $\mathcal{G}$ also a σ -field on Ω ? Justify your answer.

Solution:

To show that $\mathcal{G}$ is a σ -field on B , we must verify three properties relative to the new sample space B :

1. **Contains the sample space B :** We have $B = \Omega \cap B$. Since $\mathcal{F}$ is a σ -field on Ω , $\Omega \in \mathcal{F}$. Thus, $\Omega \cap B \in \mathcal{G}$, which means $B \in \mathcal{G}$.
2. **Closed under complementation relative to B :** Let $C \in \mathcal{G}$. By definition, $C = A \cap B$ for some $A \in \mathcal{F}$. The complement of C relative to B is:

$$B \setminus C = B \cap C^c = B \cap (A \cap B)^c = B \cap (A^c \cup B^c) = (B \cap A^c) \cup (B \cap B^c)$$

Since $B \cap B^c = \emptyset$, we have $B \setminus C = A^c \cap B$. Because $\mathcal{F}$ is a σ -field, $A^c \in \mathcal{F}$, which implies $A^c \cap B \in \mathcal{G}$.

3. **Closed under countable unions:** Let $C_1, C_2, \dots \in \mathcal{G}$. Then there exist sets $A_1, A_2, \dots \in \mathcal{F}$ such that $C_n = A_n \cap B$. We have:

$$\bigcup_{n=1}^{\infty} C_n = \bigcup_{n=1}^{\infty} (A_n \cap B) = \left(\bigcup_{n=1}^{\infty} A_n \right) \cap B$$

Since $\mathcal{F}$ is a σ -field, $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$. Thus, the union belongs to $\mathcal{G}$.

Hence, $\mathcal{G}$ is a σ -field on B .

Is $\mathcal{G}$ a σ -field on Ω ?

No, not generally. This is only true if $B = \Omega$. By definition, a σ -field on Ω must contain the set Ω itself. Since all sets in $\mathcal{G}$ are subsets of B , if B is a proper subset of Ω ($B \subsetneq \Omega$), then $\Omega \notin \mathcal{G}$. Thus, $\mathcal{G}$ is not a σ -field on Ω .

Q3. For $\Omega = \{1, 2, 3, 4\}$, find three different σ -algebras $\mathcal{F}_n$, $n = 1, 2, 3$ such that $\mathcal{F}_1 \subset \mathcal{F}_2 \subset \mathcal{F}_3$. Moreover, define a set function $\mathbb{P} : \mathcal{F}_3 \rightarrow [0, 1]$ such that $(\Omega, \mathcal{F}_3, \mathbb{P})$ becomes a probability space.

Solution:

Let $\Omega = \{1, 2, 3, 4\}$. We can define the three nested σ -algebras as follows:

- $\mathcal{F}_1 = \{\emptyset, \Omega\}$ (The trivial σ -algebra)
- $\mathcal{F}_2 = \{\emptyset, \{1\}, \{2, 3, 4\}, \Omega\}$ (Generated by the partition $\{\{1\}, \{2, 3, 4\}\}$)
- $\mathcal{F}_3 = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, \{3, 4\}, \{1, 3, 4\}, \{2, 3, 4\}, \Omega\}$
(Generated by the partition $\{\{1\}, \{2\}, \{3, 4\}\}$)

Clearly, $\mathcal{F}_1 \subset \mathcal{F}_2 \subset \mathcal{F}_3$ and all three are valid σ -algebras.

To make $(\Omega, \mathcal{F}_3, \mathbb{P})$ a probability space, we need to assign probabilities to the atomic events of $\mathcal{F}_3$. Let the partition elements have the following probabilities:

$$\mathbb{P}(\{1\}) = \frac{1}{4}, \quad \mathbb{P}(\{2\}) = \frac{1}{4}, \quad \mathbb{P}(\{3, 4\}) = \frac{1}{2}$$

For any other set $A \in \mathcal{F}_3$, its probability is the sum of the probabilities of these disjoint partition elements that make it up. This ensures $\mathbb{P}(\emptyset) = 0$, $\mathbb{P}(\Omega) = 1$, and $\mathbb{P}$ is countably additive. Thus, it is a valid probability space.

Q4. Let $A_r : r \geq 1$ be events such that $\mathbb{P}(A_r) = 1$ for all r . Show that $\mathbb{P}(\cap_{r=1}^{\infty} A_r) = 1$.

Solution:

Let $B = \cap_{r=1}^{\infty} A_r$. We wish to prove that $\mathbb{P}(B) = 1$. Consider the complement of this intersection, which by De Morgan's Laws is:

$$B^c = \left(\bigcap_{r=1}^{\infty} A_r \right)^c = \bigcup_{r=1}^{\infty} A_r^c$$

Since $\mathbb{P}(A_r) = 1$ for all $r \geq 1$, we have:

$$\mathbb{P}(A_r^c) = 1 - \mathbb{P}(A_r) = 1 - 1 = 0$$

By Boole's inequality (the union bound), the probability of a countable union of events is less than or equal to the sum of their individual probabilities:

$$\mathbb{P}(B^c) = \mathbb{P}\left(\bigcup_{r=1}^{\infty} A_r^c\right) \leq \sum_{r=1}^{\infty} \mathbb{P}(A_r^c)$$

Substituting $\mathbb{P}(A_r^c) = 0$:

$$\mathbb{P}(B^c) \leq \sum_{r=1}^{\infty} 0 = 0$$

Since probabilities must be non-negative, $\mathbb{P}(B^c) = 0$. Therefore, $\mathbb{P}(B) = 1 - \mathbb{P}(B^c) = 1 - 0 = 1$.

Q5. Let $A_1, A_2, \dots, A_N$ be a system of independent events. Assume that $\mathbb{P}(A_n) = \frac{1}{n+1}$, $n = 1, 2, \dots, N$.

- a) Find the probability that exactly one of the A_i occur?
- b) Find the probability that at most two A_i occur?

Solution:

Let $p_n = \mathbb{P}(A_n) = \frac{1}{n+1}$. Then $1 - p_n = \frac{n}{n+1}$.

First, note that the probability that *none* of the events occur is:

$$\mathbb{P}(E_0) = \prod_{j=1}^N (1 - p_j) = \prod_{j=1}^N \frac{j}{j+1} = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdots \frac{N}{N+1} = \frac{1}{N+1}$$

a) Exactly one event occurs: Let E_1 be the event that exactly one A_i occurs. Because the events are independent:

$$\mathbb{P}(E_1) = \sum_{i=1}^N \left(\mathbb{P}(A_i) \prod_{j \neq i} (1 - p_j) \right) = \sum_{i=1}^N \left(\frac{p_i}{1 - p_i} \prod_{j=1}^N (1 - p_j) \right)$$

We substitute $\prod_{j=1}^N (1 - p_j) = \frac{1}{N+1}$ and $\frac{p_i}{1 - p_i} = \frac{\frac{1}{i+1}}{\frac{i}{i+1}} = \frac{1}{i}$:

$$\mathbb{P}(E_1) = \left(\sum_{i=1}^N \frac{1}{i} \right) \cdot \frac{1}{N+1}$$

b) At most two events occur: Let E_2 be the event that exactly two events occur. Similarly:

$$\mathbb{P}(E_2) = \sum_{1 \leq i < k \leq N} \left(p_i p_k \prod_{j \neq i, k} (1 - p_j) \right) = \left(\sum_{1 \leq i < k \leq N} \frac{p_i}{1 - p_i} \frac{p_k}{1 - p_k} \right) \prod_{j=1}^N (1 - p_j)$$

$$\mathbb{P}(E_2) = \frac{1}{N+1} \sum_{1 \leq i < k \leq N} \frac{1}{ik}$$

The probability that at most two occur is $\mathbb{P}(E_0) + \mathbb{P}(E_1) + \mathbb{P}(E_2)$:

$$\mathbb{P}(\text{at most 2}) = \frac{1}{N+1} \left(1 + \sum_{i=1}^N \frac{1}{i} + \sum_{1 \leq i < k \leq N} \frac{1}{ik} \right)$$

Q6. An urn contains balls numbered from 1 to N . One ball is selected uniformly at random.

- i) Find the probability that the number on the selected ball is divisible by 3 or by 4.
- ii) Determine the limiting value of the probability obtained in part (i) as $N \rightarrow \infty$.

Solution:

i) Let A be the event that the number is divisible by 3, and B be the event that it is divisible by 4. The number of integers up to N divisible by 3 is $|A| = \lfloor N/3 \rfloor$. The number of integers up to N divisible by 4 is $|B| = \lfloor N/4 \rfloor$. The number divisible by both 3 and 4 (i.e., divisible by $\text{lcm}(3, 4) = 12$) is $|A \cap B| = \lfloor N/12 \rfloor$.

By the principle of inclusion-exclusion, the number of balls divisible by 3 or 4 is:

$$|A \cup B| = |A| + |B| - |A \cap B| = \lfloor N/3 \rfloor + \lfloor N/4 \rfloor - \lfloor N/12 \rfloor$$

Since the ball is drawn uniformly, the probability is:

$$\mathbb{P}(A \cup B) = \frac{\lfloor N/3 \rfloor + \lfloor N/4 \rfloor - \lfloor N/12 \rfloor}{N}$$

ii) For the limiting value as $N \rightarrow \infty$, note that $\frac{\lfloor x \rfloor}{x} \rightarrow 1$ as $x \rightarrow \infty$. Thus:

$$\begin{aligned} \lim_{N \rightarrow \infty} \mathbb{P}(A \cup B) &= \lim_{N \rightarrow \infty} \left(\frac{\lfloor N/3 \rfloor}{N} + \frac{\lfloor N/4 \rfloor}{N} - \frac{\lfloor N/12 \rfloor}{N} \right) \\ &= \frac{1}{3} + \frac{1}{4} - \frac{1}{12} = \frac{4+3-1}{12} = \frac{6}{12} = \frac{1}{2} \end{aligned}$$

Q7. Show that, for any events A and B with $\mathbb{P}(B) > 0$, $\mathbb{P}(A|A \cup B) \geq \mathbb{P}(A|B)$.

Solution:

By definition of conditional probability:

$$\begin{aligned} \mathbb{P}(A|A \cup B) &= \frac{\mathbb{P}(A \cap (A \cup B))}{\mathbb{P}(A \cup B)} = \frac{\mathbb{P}(A)}{\mathbb{P}(A \cup B)} \\ \mathbb{P}(A|B) &= \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} \end{aligned}$$

We wish to prove $\frac{\mathbb{P}(A)}{\mathbb{P}(A \cup B)} \geq \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$, which is equivalent to:

$$\mathbb{P}(A)\mathbb{P}(B) \geq \mathbb{P}(A \cap B)\mathbb{P}(A \cup B)$$

We know that $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$. Substituting this into the right-hand side (RHS):

$$\begin{aligned} \text{RHS} &= \mathbb{P}(A \cap B)(\mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)) \\ &= \mathbb{P}(A \cap B)\mathbb{P}(A) + \mathbb{P}(A \cap B)\mathbb{P}(B) - \mathbb{P}(A \cap B)^2 \end{aligned}$$

Now evaluate LHS – RHS:

$$\text{LHS} - \text{RHS} = \mathbb{P}(A)\mathbb{P}(B) - \mathbb{P}(A \cap B)\mathbb{P}(A) - \mathbb{P}(A \cap B)\mathbb{P}(B) + \mathbb{P}(A \cap B)^2$$

Factor by grouping:

$$\begin{aligned} &= \mathbb{P}(A)(\mathbb{P}(B) - \mathbb{P}(A \cap B)) - \mathbb{P}(A \cap B)(\mathbb{P}(B) - \mathbb{P}(A \cap B)) \\ &= (\mathbb{P}(A) - \mathbb{P}(A \cap B))(\mathbb{P}(B) - \mathbb{P}(A \cap B)) \end{aligned}$$

Since $A \cap B \subseteq A$, we have $\mathbb{P}(A) - \mathbb{P}(A \cap B) = \mathbb{P}(A \setminus B) \geq 0$. Similarly, $\mathbb{P}(B) - \mathbb{P}(A \cap B) = \mathbb{P}(B \setminus A) \geq 0$. The product of two non-negative terms is non-negative, meaning $\text{LHS} - \text{RHS} \geq 0$. Thus, $\text{LHS} \geq \text{RHS}$, and the inequality holds.

Q8. Two factories, Factory (I) and Factory (II), manufacture ballpoint pens. Factory (I) produces twice as many pens each week as Factory (II). Of the pens produced by Factory (I), 20% are defective, while 5% of the pens produced by Factory (II) are defective. A pen is selected at random from the total weekly production.

(a) What is the probability that the selected pen is not defective?

(b) If the selected pen is defective, what is the probability that it was produced by Factory I)?

Solution:

Let F_1 be the event that a pen comes from Factory I, and F_2 be Factory II. Since Factory I produces twice as many, $\mathbb{P}(F_1) = \frac{2}{3}$ and $\mathbb{P}(F_2) = \frac{1}{3}$. Let D denote the event that a pen is defective. We are given:

$$\mathbb{P}(D|F_1) = 0.20 \quad \text{and} \quad \mathbb{P}(D|F_2) = 0.05$$

(a) By the Law of Total Probability, the overall probability of a defective pen is:

$$\begin{aligned} \mathbb{P}(D) &= \mathbb{P}(D|F_1)\mathbb{P}(F_1) + \mathbb{P}(D|F_2)\mathbb{P}(F_2) \\ \mathbb{P}(D) &= 0.20 \left(\frac{2}{3}\right) + 0.05 \left(\frac{1}{3}\right) = \frac{0.40}{3} + \frac{0.05}{3} = \frac{0.45}{3} = 0.15 \end{aligned}$$

The probability that a selected pen is **not** defective is $1 - \mathbb{P}(D) = 1 - 0.15 = 0.85$.

(b) Using Bayes' Theorem, the probability that a defective pen was produced by Factory I is:

$$\mathbb{P}(F_1|D) = \frac{\mathbb{P}(D|F_1)\mathbb{P}(F_1)}{\mathbb{P}(D)} = \frac{\frac{0.40}{3}}{0.15} = \frac{0.40}{0.45} = \frac{40}{45} = \frac{8}{9}$$

Q9. On flights from Delhi to Kolkata, 90% depart on time and arrive on time, 6% depart on time but arrive late, 1% depart late but arrive on time, and 3% depart late and arrive late. What is the probability that, given a flight late, it will arrive on time?

Solution:

Let's define the events:

- D_T : Depart on time, D_L : Depart late
- A_T : Arrive on time, A_L : Arrive late

The given probabilities are:

$$\mathbb{P}(D_T \cap A_T) = 0.90, \quad \mathbb{P}(D_T \cap A_L) = 0.06$$

$$\mathbb{P}(D_L \cap A_T) = 0.01, \quad \mathbb{P}(D_L \cap A_L) = 0.03$$

Note that they sum to 1.00. The phrase "given a flight late" refers to the flight *departing* late (since checking the likelihood of it arriving on time given it arrived late would be nonsensical). Thus, we want $\mathbb{P}(A_T|D_L)$. First, find $\mathbb{P}(D_L)$ (marginal probability of departing late):

$$\mathbb{P}(D_L) = \mathbb{P}(D_L \cap A_T) + \mathbb{P}(D_L \cap A_L) = 0.01 + 0.03 = 0.04$$

Now compute the conditional probability:

$$\mathbb{P}(A_T|D_L) = \frac{\mathbb{P}(A_T \cap D_L)}{\mathbb{P}(D_L)} = \frac{0.01}{0.04} = \frac{1}{4} = 0.25$$

Q10. A biased coin, with probability $\frac{1}{3}$ of head turning up, is tossed repeatedly. Let p_n be the probability that even number of heads has occurred after n tosses (zero is an even number). Show that $p_0 = 1$, $p_n = \frac{1}{3}(1 - p_{n-1}) + \frac{2}{3}p_{n-1}$ for $n \geq 1$. Solve this difference equation.

Solution:

Before any tosses ($n = 0$), 0 heads have appeared. Since 0 is an even number, the event holds with certainty, hence $p_0 = 1$.

For $n \geq 1$, we can achieve an even number of heads after n tosses in two mutually exclusive ways:

1. We had an even number of heads after $n - 1$ tosses (probability p_{n-1}), and the n -th toss is a Tail (probability $\frac{2}{3}$).
2. We had an odd number of heads after $n - 1$ tosses (probability $1 - p_{n-1}$), and the n -th toss is a Head (probability $\frac{1}{3}$).

Summing these probabilities gives the recurrence relation:

$$p_n = \frac{2}{3}p_{n-1} + \frac{1}{3}(1 - p_{n-1}) = \frac{1}{3}(1 - p_{n-1}) + \frac{2}{3}p_{n-1}$$

Simplifying the expression:

$$p_n = \frac{1}{3} + \frac{1}{3}p_{n-1}$$

To solve this linear difference equation, subtract $\frac{1}{2}$ from both sides (this is found by solving for the steady-state $p = 1/3 + 1/3p \implies p = 1/2$):

$$p_n - \frac{1}{2} = \frac{1}{3}p_{n-1} - \frac{1}{6} = \frac{1}{3}\left(p_{n-1} - \frac{1}{2}\right)$$

This forms a geometric sequence where the term $(p_n - \frac{1}{2})$ is multiplied by $\frac{1}{3}$ at each step. Thus:

$$p_n - \frac{1}{2} = \left(\frac{1}{3}\right)^n \left(p_0 - \frac{1}{2}\right)$$

Since $p_0 = 1$, $p_0 - \frac{1}{2} = \frac{1}{2}$. We get:

$$p_n = \frac{1}{2} + \frac{1}{2}\left(\frac{1}{3}\right)^n = \frac{1}{2}(1 + 3^{-n})$$

Q11. There are two piles of playing cards. The first pile contains 8 cards: 4 aces, 2 kings, and 2 queens. The second pile also contains 8 cards: 1 ace, 4 kings, and 3 queens. A card is selected at random from the first pile and placed into the second pile. The second pile is then shuffled, and one card is drawn at random from it. If the card drawn from the second pile is an ace, what is the probability that the card transferred from the first pile was also an ace?

Solution:

Let T_A, T_K, T_Q be the events that an Ace, King, or Queen is transferred from the first pile. Their probabilities are:

$$\mathbb{P}(T_A) = \frac{4}{8} = \frac{1}{2}, \quad \mathbb{P}(T_K) = \frac{2}{8} = \frac{1}{4}, \quad \mathbb{P}(T_Q) = \frac{2}{8} = \frac{1}{4}$$

Let D_A be the event that an Ace is drawn from the second pile. The second pile now has 9 cards. The probability of D_A conditioned on the transfer is:

- If T_A , 2nd pile has 2 Aces out of 9: $\mathbb{P}(D_A|T_A) = \frac{2}{9}$

- If T_K , 2nd pile has 1 Ace out of 9: $\mathbb{P}(D_A|T_K) = \frac{1}{9}$
- If T_Q , 2nd pile has 1 Ace out of 9: $\mathbb{P}(D_A|T_Q) = \frac{1}{9}$

By Bayes' theorem, we want $\mathbb{P}(T_A|D_A)$:

$$\begin{aligned}\mathbb{P}(T_A|D_A) &= \frac{\mathbb{P}(D_A|T_A)\mathbb{P}(T_A)}{\mathbb{P}(D_A|T_A)\mathbb{P}(T_A) + \mathbb{P}(D_A|T_K)\mathbb{P}(T_K) + \mathbb{P}(D_A|T_Q)\mathbb{P}(T_Q)} \\ &= \frac{\left(\frac{2}{9}\right)\left(\frac{1}{2}\right)}{\left(\frac{2}{9}\right)\left(\frac{1}{2}\right) + \left(\frac{1}{9}\right)\left(\frac{1}{4}\right) + \left(\frac{1}{9}\right)\left(\frac{1}{4}\right)} = \frac{\frac{2}{18}}{\frac{2}{18} + \frac{1}{36} + \frac{1}{36}} = \frac{\frac{4}{36}}{\frac{4}{36} + \frac{1}{36} + \frac{1}{36}} = \frac{4}{6} = \frac{2}{3}\end{aligned}$$

Q12. The coefficients a , b , and c of the quadratic equation $ax^2 + bx + c = 0$ are obtained by rolling a fair six-sided die three times, with the outcomes taken as the values of a , b , and c , respectively. What is the probability that the quadratic equation has two equal real roots?

Solution:

For a quadratic equation to have two equal real roots, its discriminant must be zero:

$$b^2 - 4ac = 0 \implies b^2 = 4ac$$

The dice outcomes imply $a, b, c \in \{1, 2, 3, 4, 5, 6\}$. Since $4ac$ is an even number, b^2 must be even, so b must be even ($b \in \{2, 4, 6\}$).

- Case $b = 2$: $b^2 = 4 \implies 4ac = 4 \implies ac = 1$. The only combination is $(a = 1, c = 1)$. (**1 solution**)
- Case $b = 4$: $b^2 = 16 \implies 4ac = 16 \implies ac = 4$. The valid combinations from the die are $(1, 4), (2, 2), (4, 1)$. (**3 solutions**)
- Case $b = 6$: $b^2 = 36 \implies 4ac = 36 \implies ac = 9$. The only valid combination is $(3, 3)$. (**1 solution**)

The total number of favorable outcomes is $1 + 3 + 1 = 5$. The total number of possible outcomes for rolling three dice is $6 \times 6 \times 6 = 216$. Therefore, the probability is $\frac{5}{216}$.

Q13. Let A_1 , A_2 and A_3 be independent events. Prove that $\mathbb{P}(A_1 \cup A_2 \cup A_3) = 1 - \prod_{i=1}^3 (1 - \mathbb{P}(A_i))$.

Solution:

We can evaluate the probability of the union by utilizing the probability of its complement. By De Morgan's Laws, the complement of the union of events is the intersection of their complements:

$$(A_1 \cup A_2 \cup A_3)^c = A_1^c \cap A_2^c \cap A_3^c$$

Since the events A_1, A_2, A_3 are independent, their complements A_1^c, A_2^c, A_3^c are also independent. For independent events, the probability of their intersection is the product of their probabilities:

$$\mathbb{P}(A_1^c \cap A_2^c \cap A_3^c) = \mathbb{P}(A_1^c)\mathbb{P}(A_2^c)\mathbb{P}(A_3^c)$$

The probability of a complement is $\mathbb{P}(A_i^c) = 1 - \mathbb{P}(A_i)$. Thus:

$$\mathbb{P}((A_1 \cup A_2 \cup A_3)^c) = \prod_{i=1}^3 (1 - \mathbb{P}(A_i))$$

The probability of an event is 1 minus the probability of its complement:

$$\mathbb{P}(A_1 \cup A_2 \cup A_3) = 1 - \mathbb{P}((A_1 \cup A_2 \cup A_3)^c) = 1 - \prod_{i=1}^3 (1 - \mathbb{P}(A_i))$$

This completes the proof.

Q14. Let A and B be independent events. Prove or disprove that A and B^c , as well as A^c and B^c are independent.

Solution:

We are given that A and B are independent, so $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.

1. Proving A and B^c are independent: The event A can be partitioned into two disjoint subsets: $A = (A \cap B) \cup (A \cap B^c)$. By the additivity axiom:

$$\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c)$$

$$\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B)$$

Substituting the independence of A and B :

$$\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A)\mathbb{P}(B) = \mathbb{P}(A)(1 - \mathbb{P}(B))$$

Since $1 - \mathbb{P}(B) = \mathbb{P}(B^c)$, we get $\mathbb{P}(A \cap B^c) = \mathbb{P}(A)\mathbb{P}(B^c)$. This proves A and B^c are independent.

2. Proving A^c and B^c are independent: Following a similar logic, use the fact that $B^c = (A \cap B^c) \cup (A^c \cap B^c)$.

$$\mathbb{P}(A^c \cap B^c) = \mathbb{P}(B^c) - \mathbb{P}(A \cap B^c)$$

Substituting the newly proven result $\mathbb{P}(A \cap B^c) = \mathbb{P}(A)\mathbb{P}(B^c)$:

$$\mathbb{P}(A^c \cap B^c) = \mathbb{P}(B^c) - \mathbb{P}(A)\mathbb{P}(B^c) = \mathbb{P}(B^c)(1 - \mathbb{P}(A))$$

Since $1 - \mathbb{P}(A) = \mathbb{P}(A^c)$, we get $\mathbb{P}(A^c \cap B^c) = \mathbb{P}(A^c)\mathbb{P}(B^c)$. Thus, A^c and B^c are independent.

Q15. A fair die is rolled five times. Let A_{ij} be the event that the i -th and j -th rolls show the same number, where $1 \leq i < j \leq 5$. Prove that the events $\{A_{ij} : 1 \leq i < j \leq 5\}$ are pairwise independent but not independent.

Solution:

Let $X_1, X_2, \dots, X_5$ be the independent outcomes of the 5 rolls, uniformly distributed over $\{1, 2, 3, 4, 5, 6\}$. The event A_{ij} occurs if $X_i = X_j$. There are 6 ways they match, out of $6^2 = 36$ pairs, so:

$$\mathbb{P}(A_{ij}) = \frac{6}{36} = \frac{1}{6} \quad \text{for all } i < j$$

Pairwise Independence: Consider two distinct events A_{ij} and A_{kl} . *Case 1 (Disjoint indices):* If $\{i, j\} \cap \{k, l\} = \emptyset$ (e.g., A_{12} and A_{34}). The events depend on disjoint subsets of independent dice rolls, hence they are strictly independent:

$$\mathbb{P}(A_{12} \cap A_{34}) = \mathbb{P}(A_{12})\mathbb{P}(A_{34}) = \left(\frac{1}{6}\right)^2 = \frac{1}{36}$$

Case 2 (One shared index): If $\{i, j\} \cap \{k, l\}$ has size 1 (e.g., A_{12} and A_{13}). The intersection implies $X_1 = X_2$ and $X_1 = X_3$, meaning all three rolls are identical. There are 6 such combinations out of $6^3 = 216$ outcomes for the 3 dice.

$$\mathbb{P}(A_{12} \cap A_{13}) = \frac{6}{216} = \frac{1}{36}$$

In both cases, $\mathbb{P}(A_{ij} \cap A_{kl}) = \mathbb{P}(A_{ij})\mathbb{P}(A_{kl}) = \frac{1}{36}$. Thus, they are pairwise independent.

Not Mutually Independent: Consider the three events A_{12}, A_{23}, A_{13} . The intersection $A_{12} \cap A_{23} \cap A_{13}$ occurs when $X_1 = X_2 = X_3$. As established above, this probability is $\frac{1}{36}$. However, the product of their individual probabilities is:

$$\mathbb{P}(A_{12})\mathbb{P}(A_{23})\mathbb{P}(A_{13}) = \left(\frac{1}{6}\right)^3 = \frac{1}{216}$$

Since $\frac{1}{36} \neq \frac{1}{216}$, the events are not mutually independent.

Q16. Let $\Omega = \{1, 2, 3, \dots, 7\}$, $\mathcal{F}$ be the power set of Ω , and $\mathbb{P}(A) = \frac{|A|}{7}$ for all $A \in \mathcal{F}$. Show that, if A and B are independent events, then at least one of A and B is either empty set or Ω .

Solution:

Let $a = |A|$ and $b = |B|$, so $\mathbb{P}(A) = \frac{a}{7}$ and $\mathbb{P}(B) = \frac{b}{7}$. Notice $0 \leq a \leq 7$ and $0 \leq b \leq 7$. Let $c = |A \cap B|$, so $\mathbb{P}(A \cap B) = \frac{c}{7}$, where $0 \leq c \leq 7$.

If A and B are independent, then:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) \implies \frac{c}{7} = \left(\frac{a}{7}\right)\left(\frac{b}{7}\right) \implies 7c = ab$$

Since 7 is a prime number, Euclid's lemma dictates that if a prime divides a product of two integers (ab), it must divide at least one of those integers. Thus, 7 must divide a or 7 must divide b .

The possible values for a and b are integers from 0 to 7. The only multiples of 7 in this range are 0 and 7. Therefore:

- Either $a = 0 \implies |A| = 0 \implies A = \emptyset$
- Or $a = 7 \implies |A| = 7 \implies A = \Omega$
- Or $b = 0 \implies |B| = 0 \implies B = \emptyset$
- Or $b = 7 \implies |B| = 7 \implies B = \Omega$

Thus, at least one of the sets A or B must be trivially empty ($\emptyset$) or the entire sample space (Ω).

Q17. Two fair six-sided dice are rolled. Show that the event that the sum of the numbers shown is 7 is independent of the number shown on the first die.

Solution:

Let S_7 be the event that the sum of the dice is 7. The favorable outcomes are $\{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$. There are 6 such outcomes out of a total 36.

$$\mathbb{P}(S_7) = \frac{6}{36} = \frac{1}{6}$$

Let D_x be the event that the first die shows the value $x \in \{1, 2, 3, 4, 5, 6\}$.

$$\mathbb{P}(D_x) = \frac{1}{6}$$

The intersection $S_7 \cap D_x$ represents the event where the first die is x and the sum is 7. This completely dictates the value of the second die, which must be $7 - x$. There is exactly 1 such outcome (namely $(x, 7 - x)$) for any given x .

$$\mathbb{P}(S_7 \cap D_x) = \frac{1}{36}$$

Now we test for independence:

$$\mathbb{P}(S_7)\mathbb{P}(D_x) = \left(\frac{1}{6}\right)\left(\frac{1}{6}\right) = \frac{1}{36} = \mathbb{P}(S_7 \cap D_x)$$

Since $\mathbb{P}(S_7 \cap D_x) = \mathbb{P}(S_7)\mathbb{P}(D_x)$ holds for any outcome x of the first die, the event that the sum is 7 is perfectly independent of the outcome of the first die.